Box 1: Metrics of (In)determinacy

To advance our understanding of growth strategies under climate change, we need not only improved methods for quantifying the degree of (in)determinacy—but also more empirical data to apply and validate these approaches across species and ecosystems. Moving beyond a simple binary classification, we propose a set of metrics that span scales, from organ-level traits to landscape-scale proxies:

Ratio of preformed to neoformed organs: The ratio of the number of leaves (or leaf scars) at the end of the season to the number of leaf primordia initially formed in buds provides a direct measure of (in)determinacy. Values greater than one indicate a higher degree of indeterminate growth. First described over a century ago (Moore, 1909), this metric has been used in only a few studies (Damascos *et al.*, 2005; Kikuzawa, 1983; Guédon *et al.*, 2006). Although quite distinct and easily counted in some species (e.g. maples; Critchfield 1971), there are only slight morphological differences between neoformed and preformed leaves in others (e.g., black cottonwood). Time-lapse imaging, labeling or artificial wounding can facilitate the distinction between preformed and neoformed organs. The number of leaf primordia can be determined through bud dissection, but this precludes the counting of neoformed leaves that would later arise from the same meristem. Micro-CT imaging may offer a modern, non-destructive alternative to bud dissection (Kovaleski *et al.*, 2019).

Seasonal growth patterns: Temporal tracking of shoot elongation and wood formation—via micro-coring, dendrometers, or regular shoot measurements—reveals when and how new tissue is produced. Wood anatomical analyses distinguish cell formation from later development, especially useful in combination with artificial wounding of the cambium that acts as a 'marker in time' (pinning technique; Gärtner & Farahat 2021). High-resolution LiDAR and photogrammetry are increasingly capable of capturing shoot elongation and structural changes across whole crowns or stands (Zhang *et al.*, 2019; Jin *et al.*, 2022).

Seasonal trends in canopy greenness: Satellite and drone imagery can track vegetation green-up and senescence at large spatial scales. For example, Meng *et al.* (2024) found that time allocation between green-up and senescence remained consistent across temperate ecosystems, despite substantial variation in growing season length and warming intensity. Such temporal dynamics of green coloration 1m3ay provide indirect insights into determinacy as it reflects how primary growth is distributed across the season — more extended or bimodal green coloration patterns may indicate indeterminate growth. However, green coloration is a crude proxy and may confound photosynthetic activity with actual organogenesis or elongation, so its interpretation as a (in)determinacy measure should be treated with caution